

Wireless Networks

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Content



- Preliminaries
- Components
- Types of wireless networks. Characteristics.
- Mobile IP
- Applications
- Wireless Sensor Networks (WSN)

Preliminaries

- The wireless signal is transported through air by electromagnetic waves
- The radio frequency spectrum is a continuum of electromagnetic waves used for data and voice communications
- The networks which transmit signals through the air, through radio frequency, (RF) are known as wireless networks or WLANs

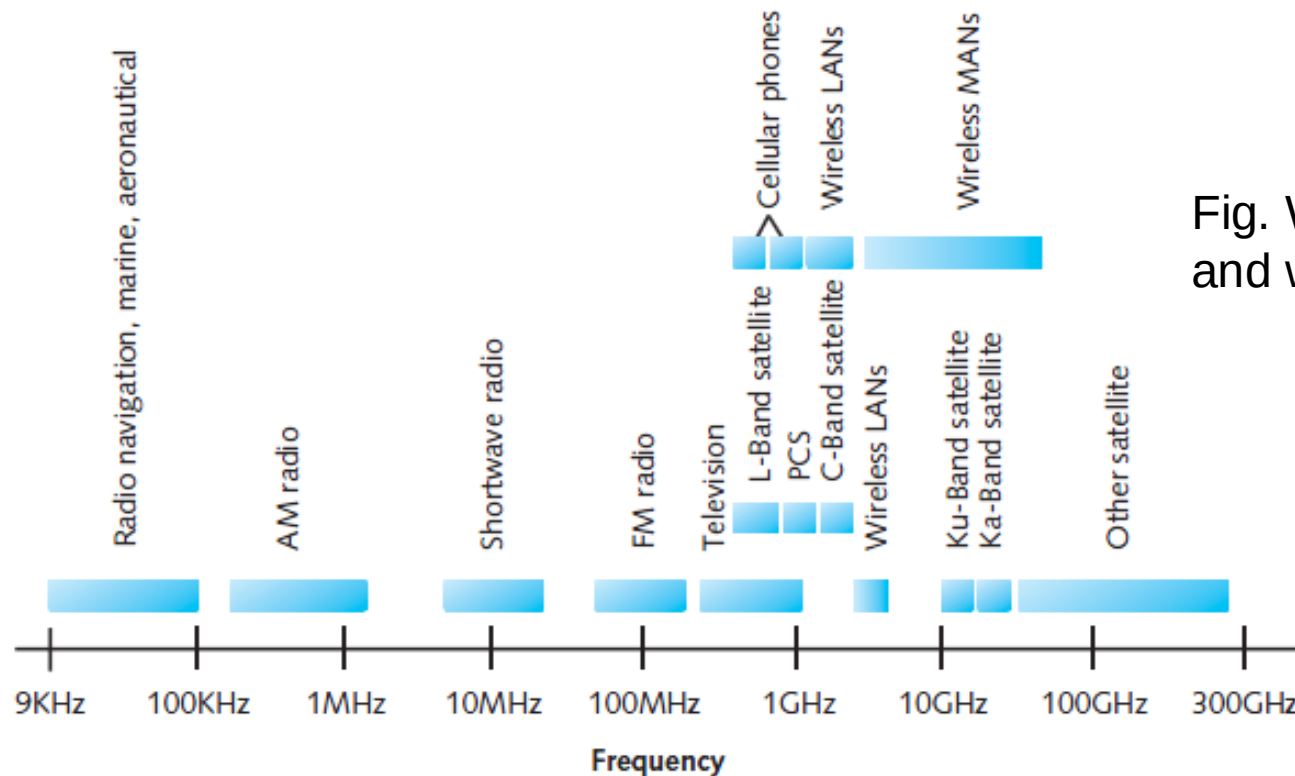


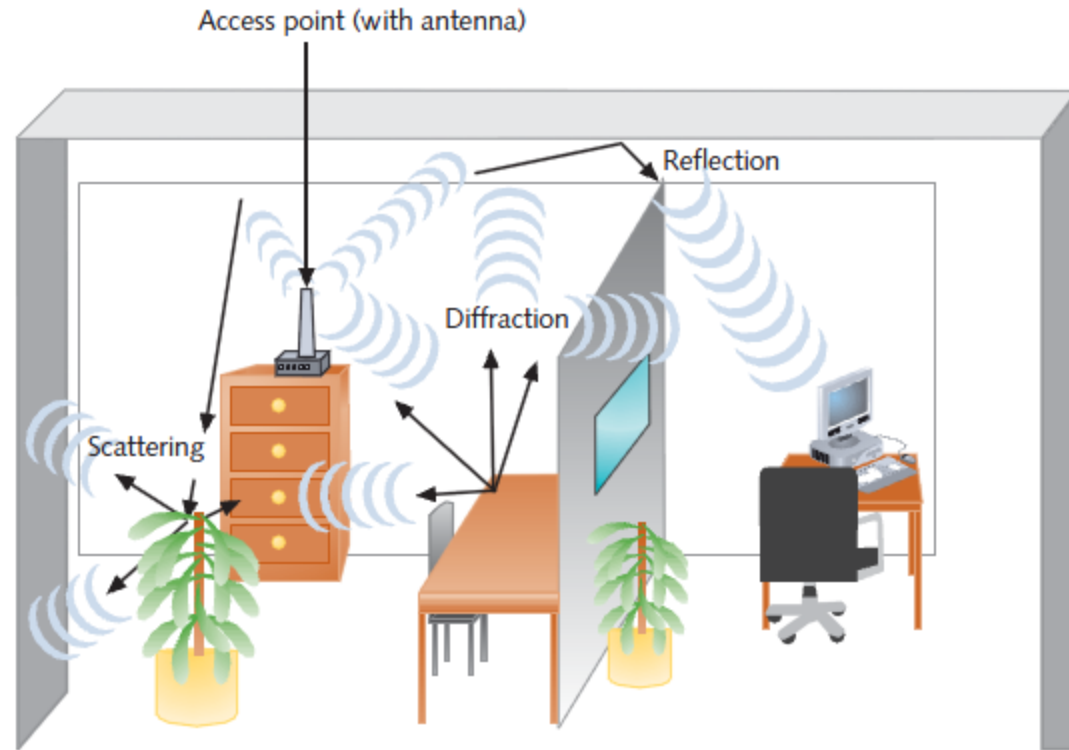
Fig. Wireless spectrum and wireless services

[Network+ Guide to Networks,
Tamara Dean]

Preliminaries

Wireless signal propagation:

- ideally: in straight line from a sender to a receiver - LOS (*line-of-sight*)
- *reflection*
- *diffraction* – caused by obstacles like objects with sharp edges (e.g., corners of desks and walls)
- *scattering* – caused by the encounter with an object of small size compared with the signal wavelength or can be determined by the roughness of a surface; for outdoor signals, the rain, fog, or snow can cause this effect



[Network+ Guide to Networks,
Tamara Dean]

Preliminaries

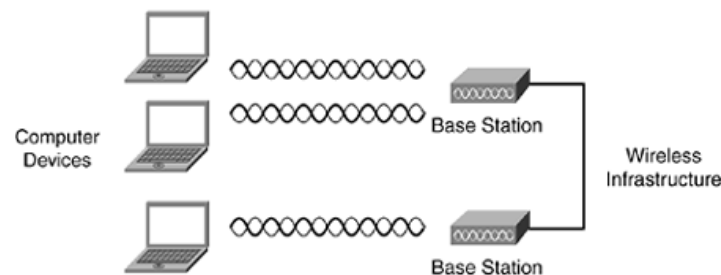
- *Fixed versus Mobile*
 - *fixed* wireless systems – sender's and receiver's locations do not change (*point-to-point* link)
 - *The entire energy is used for transmitting the signal in a single direction and not for transmitting in a large area*
 - *mobile* wireless systems – the receiver can be anywhere in the sender's covered area

Preliminaries

- Problems:
 - Coverage and penetration
 - Bandwidth
 - Latency
 - Transmission reliability
 - Standardization
- Challenges:
 - Location detection
 - Movement detection and communication path (re)establishment

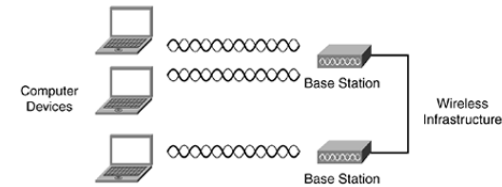
Preliminaries

- In wireless networks the communication is realized between devices: smartphones/PDAs (Personal Digital Assistants), laptops, tablets, servers, printers, sensors, ...
 - Characteristics of these devices
 - Processor, memory
 - Interface with a particular type of network
 - Obs.: In the past, traditional phones did not belong to this category, but the majority of the ones existing in the present incorporate the above features



[Wireless Networks first-step,
Jim Geier]

Wireless Networks | Components



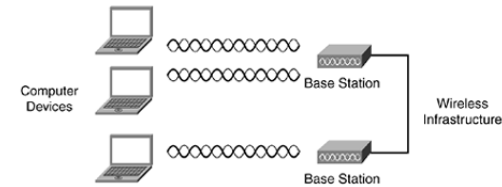
- **Devices**

Aspects of interest

- Size & Weight
- Available memory for applications and data
- Processor speed
- Characteristics of the display (resolution, color depth, etc.)
- Input mechanisms (data acquisition)
- Support for mobility from the operating system



Wireless Networks | Components



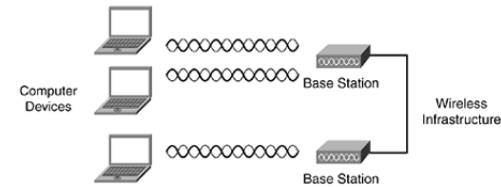
- **Devices**

Aspects of interest

- Slots for future extensions
(additional memory, connectivity with other devices)
- Battery life
- Integrated characteristics: digital camera, keyboard, infrared ports, Bluetooth, etc.
- Software support: applications, development tools, mobile browsers, hardware drivers, etc.



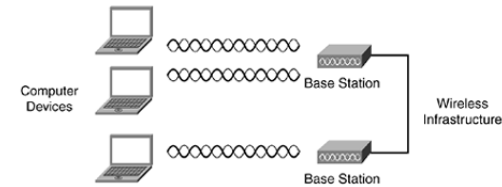
Wireless Networks | Components



- **NIC (Network Interface Card)**
 - Provides the interface between the devices and the wireless network infrastructure
- Implements a certain standard (e.g., IEEE 802.11ad) => allows interfacing with a compatible wireless network



Wireless Networks | Components



In a wireless network infrastructure, the following components enable the communication or provide a series of functionalities:

- ***Base stations***

- Components which interface wireless communications with wired ones
- Example: access points or routers
- Can support *point-to-point* or *point-to-multipoint* communication



Wireless Networks | Components

- ***Access controllers***

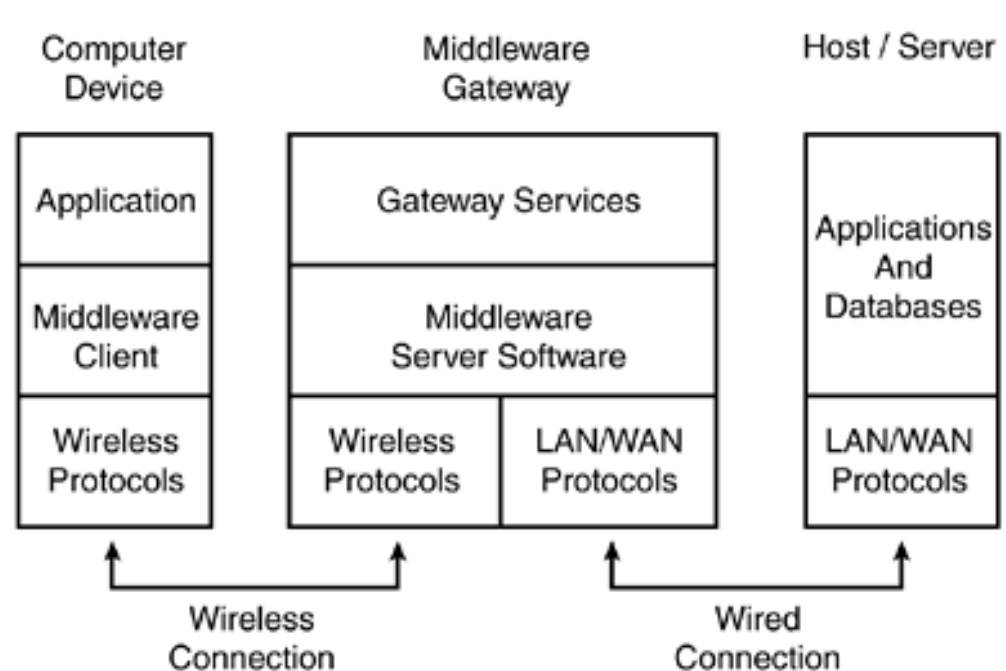
- Usually it is a hardware component which stands between the access point and the protected network
- Functionalities: user authentication and authorization, encryption, bandwidth management
- There are access points which integrate those functionalities

Wireless Networks | Components

- Connection software

- Its purpose is to ensure an efficient and secure communication in the wireless network

- optimization
- smart reboot
- efficient discovery
- etc.



Wireless Networks

- Categories – depending on the size of the area that is covered
 - **WPAN** (*Wireless Personal-Area Network*)
 - **WLAN** (*Wireless Local-Area Network*)
 - **WMAN** (*Wireless Metropolitan-Area Network*)
 - **WWAN** (*Wireless Wide-Area Network*)

Wireless Networks | WPAN

WPAN (*Wireless Personal-Area Network*)

- Small operating space ($\sim 10\text{m}$, 20m)
- Replaces interconnection cables to other equipment
- Performance: moderate
- Standards: IrDa, Bluetooth, (IEEE) 802.15, 802.11ah/Wi-Fi HaLow (IoT) etc.



Wireless Networks | WPAN



WPAN connectivity standards

- **IrDa** (*Infrared Data Association*): bidirectional point-to-point communication via infrared ports
 - Can be used in a limited area without obstacles (*line-of-sight*)
 - It is not affected by *RF* (*radio frequency*) interferences

Exemple: *smartphone* – *PC* synchronization

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WPAN connectivity standards

- **Bluetooth**

- Introduced in 1998, ensures interconnectivity between devices, using 2.4 GHz frequency
- A device can be configured to be discoverable generally, limited, or not at all (the discovery is automatic)
- 8 devices form a *piconet* (1 *master* si 7 *slave* devices). The formed *ad-hoc* network has a maximum covered surface of 10m (usual scenario; the range depends on the power level (Bluetooth Class), newer versions increasing the range in certain conditions)
 - Piconets are dynamically established depending on the way the devices get in or out of a certain area

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WPAN connectivity standards

- **Bluetooth**

- *Piconet master* is a device whose characteristics (clock, address) define the characteristics of the physical channel of the piconet
- At a certain time, data can be transferred between a master and a slave; the master device uses a *round-robin* mechanism for communicating with each slave device
- Any device can go from a slave state in a master slave and vice versa – see www.bluetooth.com
- Many *piconets* = *scatternet*

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WPAN connectivity standards

IEEE 802.15: based on Bluetooth model, to offer wireless communication standards

www.ieee802.org/15

- Also offers interoperability with devices that support 802.11
- 802.15.1 – bandwidth: 1 Mbps
- 802.15.3 – bandwidth: between 11 and 55 Mbps
- And many other versions, with different characteristics and applicability

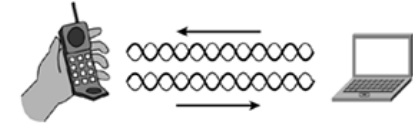
UWB (Ultra Wide Band) – radio technology for communications in limited areas, using >5Ghz frequencies

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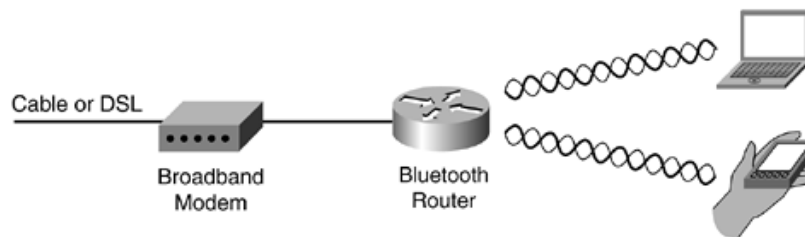


Uses of **PAN** wireless systems:

- **Synchronize**
 - Ex.: Between mobile phones and a laptop, tablet, or PC
- **Multimedia streaming**
- **Control**
 - Ex.: wireless mouse wireless, wireless keyboard
- **Easier access to devices**
 - Ex.: wireless connection between a PC and a printer
- **Internet Connection**



- **Enterprise**
 - Use WPAN for synchronizations and device access
 - For Internet connectivity, WLAN is used



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Uses of PAN wireless systems:

- **Bluetooth**



'Smart Hip' uses *Bluetooth* technology to monitor joint implants

When it comes to replacing wires with *Bluetooth* wireless technology, we can't think of any better place to do this than with medical devices surgically implanted in your body. That's why the new "Smart Hip" sounds so intriguing.

The Smart Hip monitors the performance of hip implants in real time, sending the information wirelessly from inside your body to a nearby *Bluetooth* enabled computer. The Smart Hip is actually a network of capsules, measuring sensors and actuators placed on the hip implant. Once activated by the doctor, these sensors send information to help prevent eventual problems after surgery. This is important because of the 1 million hip joint surgeries annually in Europe and the United States, an estimated that 5% to 10% of them eventually generate further health problems, which usually requires additional surgery.

The Smart Hip is being developed in Portugal by researchers at the University of Porto and other organizations.

<http://www.bluetooth.com/English/Experience/Pages/cool-clever.aspx>

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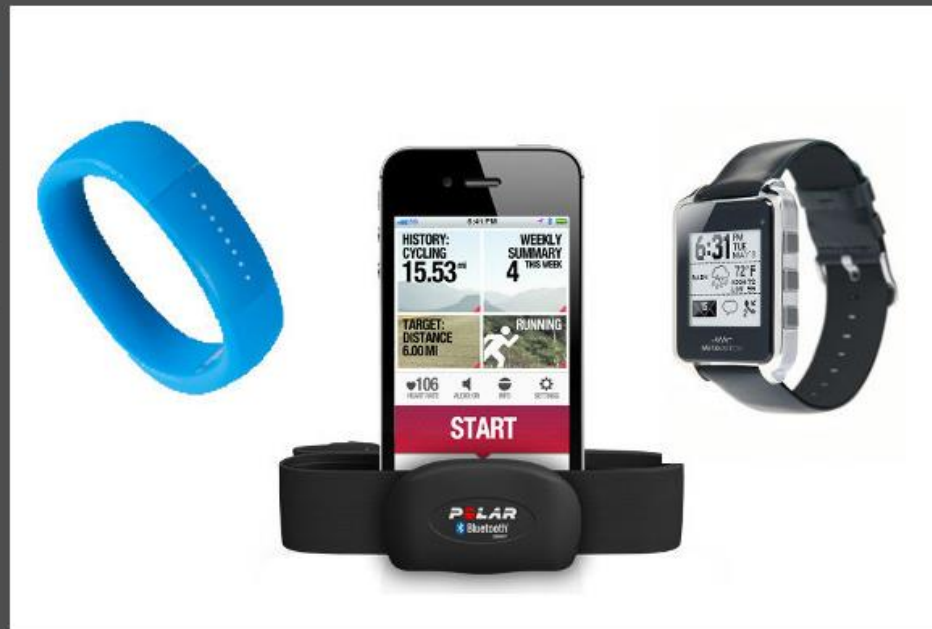
WPAN connectivity standards

- **Bluetooth**

GADGETS Bluetooth , CES ,

Bluetooth powers gizmos for health, wristwatches, and more

Yardena Arar @dennyarar Jan 8, 2013 12:52 PM



Bluetooth for all

LAS VEGAS— Forget mice and headsets: The latest Bluetooth gadgets can help monitor your health, keep track of personal belongings, and empower a wristwatch to show text messages and the identity of smartphone callers.

Most of the devices here at CES and on display at an event held for Bluetooth device makers use the low-voltage wireless technology to communicate with mobile apps to either record or deliver information and create peer-to-peer wireless connections.

At the Bluetooth SIG's CES event, vendors showed a slew of electronic nagging gadgets, including a toothbrush for monitoring dental hygiene and wristwatches that can deliver text messages and caller ID info from a paired cellphone.

Tweet 114 Like 30 +1 3 Share

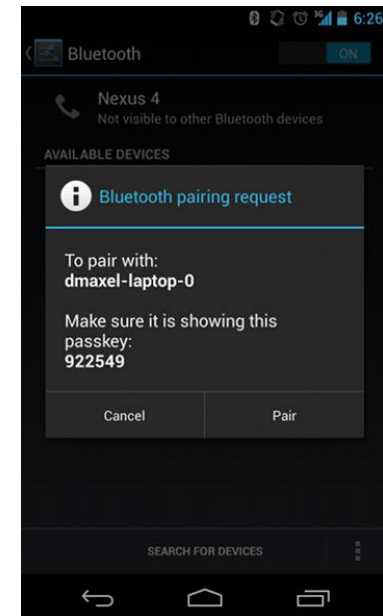
<http://www.pcworld.com/article/2024238/bluetooth-powers-gizmos-for-health-wristwatches-and-more.html>

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WPAN connectivity standards

- **Bluetooth**



<http://www.makeuseof.com/tag/top-uses-for-bluetooth-on-your-android-phone/>

Wireless Networks | WLAN

WLAN (*Wireless Local-Area Network*)

- Operating space ~ 100m (buildings, campuses)
- Extension/alternative to conventional LANs
- Performance: high
- Standards for physical and data link layers:
 - 802.11b, a, g, n, WiGig (802.11ad, 802.11ay)
 - (802.11k, r, y, n, w, p, z, v, u, s) => IEEE 802.11-2012
 - 802.11ac, Wi-Fi 6(E) (802.11ax), Wi-Fi 7 (802.11be)
 - HiperLAN/2 (discontinued)



[Wireless Networks first-step,
Jim Geier]

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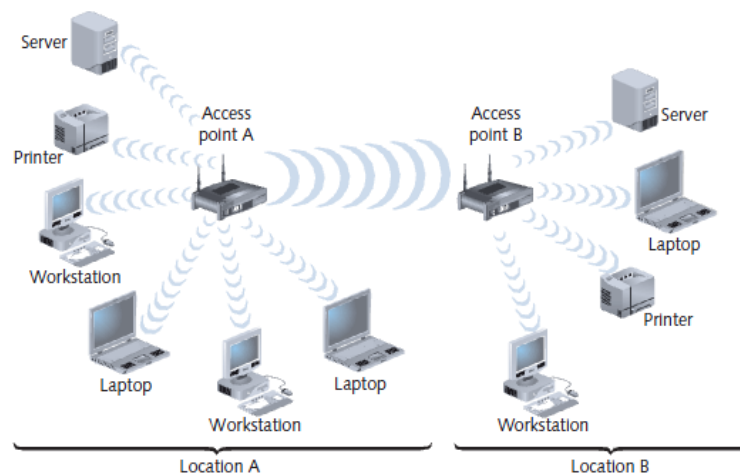
WLAN – Components

- **User device**
 - PCs, laptops, PDAs, smartphones -> properly equipped
- **Radio NIC** (*Network Interface Card*) or *adapters* or *radio cards*
 - Operate inside the device and offers wireless connectivity
 - Implements and supports one or many versions of a standard (e.g., 802.11a, 802.11b/g,)
- **Access point**
 - Contains a radio card which communicates with a user device through a wireless network

Wireless Networks | WLAN

WLAN – Components

- Example of an **access point** which offers the possibility to connect a WLAN to a LAN:
 - offers the possibility to configure various functionalities (e.g., through a web interface)
 - ✓ SSID (Service Set Identifier)
 - ✓ transmission power
 - ✓ an identifier for the channel (e.g., 1, 6, 11)
 - ✓ establishing a security level – activating an encryption mechanism (e.g., WPA2, ...)



LAN networks
interconnection

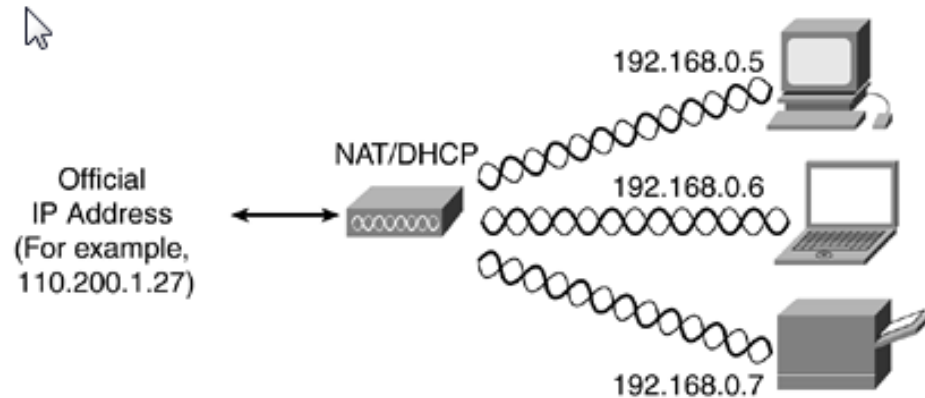
[Network+ Guide to Networks,
Tamara Dean]

Wireless Networks | WLAN

WLAN – Components

Routers

- Device which ensures packet routing (using IP, routing tables and other internal protocols ... see courses 2, 3, and 12)
- Wireless router – adds to an Ethernet router the functionality provided by an access point (802.11)
- The routers implement NAT (Network Address Translation) and DHCP (Dynamic Host Configuration Protocol)

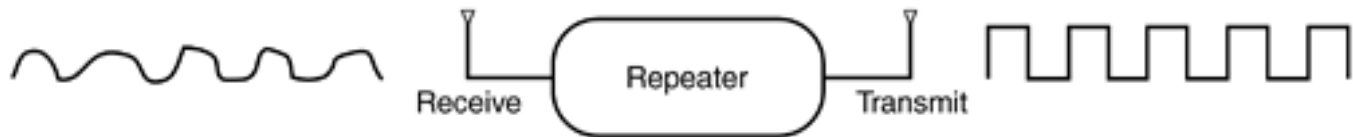


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WLAN – Components

Repeater

- Devices which ensure signal regeneration (=>increase the covered area) without adding new access points



- Reduce the performance of WLANs; the repeater receives and retransmits each *frame* on the same radio channel, which doubles the amount of traffic in the network
- Obs.: The repeaters must be optimally used in order to avoid duplicating the transmitted data

Wireless Networks | WLAN

WLAN systems

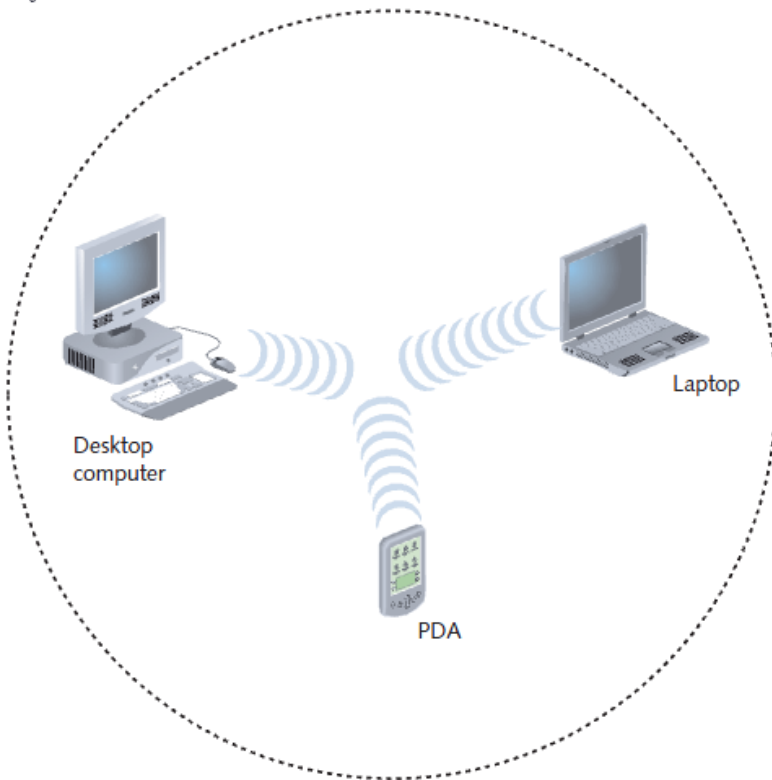


Fig. **Ad-Hoc WLAN**

- Small number of nodes
- Direct transmission between devices via wireless NIC
- APs are not used
- Also called *peer-to-peer* networks

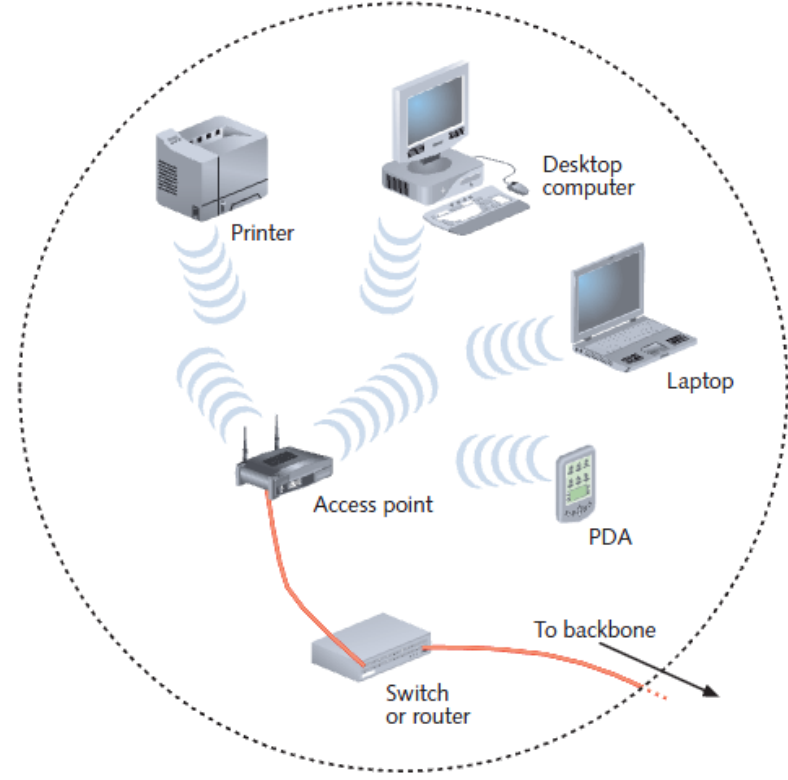


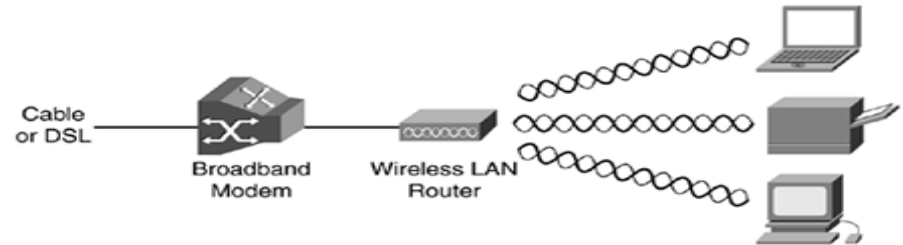
Fig. **Infrastructure WLAN**

- APs are used

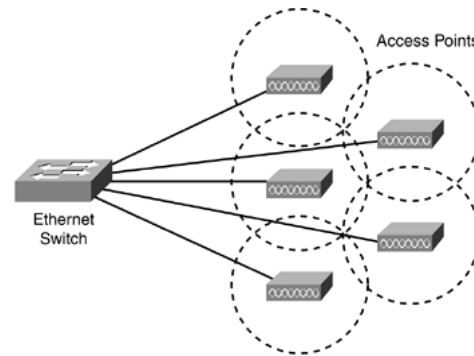
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WLAN systems

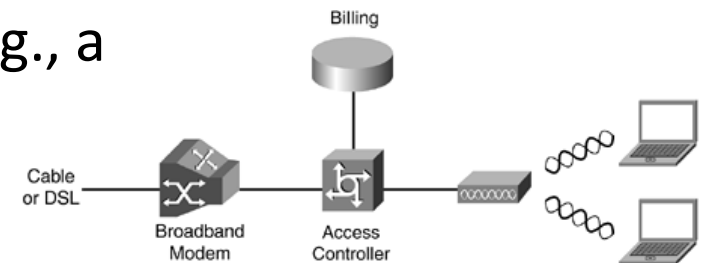
- for "*Home&Small Office*"



- for Enterprise



- **Public WLANs** – can have a simpler (e.g., a wireless router) or more complex configuration



<https://www.lifewire.com/how-to-find-free-wifi-locations-1358040>

[Wireless Networks first-step,
Jim Geier]

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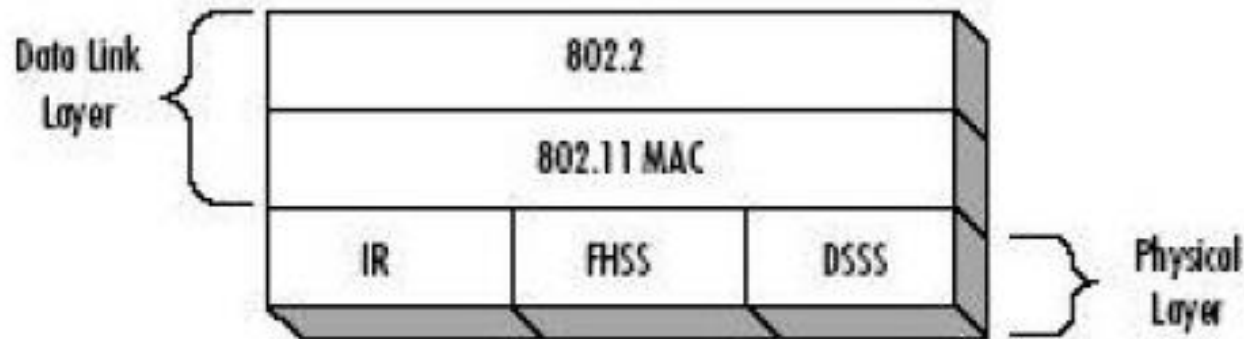
WLAN connectivity standards

- Wi-Fi Alliance – organization which proposed the grouping of all existing and future 802.11 standards under **WI-FI**
<https://www.wi-fi.org>
Characteristics: used frequency, covered area, ...
- *Use case*: you have a laptop with a NIC wireless which supports one or many 802.11 standards and you walk in the area of a *hot spot*
 - *Association* – an action of the MAC sublevel which involves the exchange of packets between AP and device, after which the user is notified about the existence of a wireless network
 - active scan – the device sends a special frame (probe) on all available channels in his area; an AP could respond with a status code and an ID for the respective station
 - passive scan – the device listens on all channels for identifying a *beacon frame*, which contains the transmission rate, SSID (service set identifier) (e.g., NetSpot - wireless network monitor available for Mac OS, Vistumbler for Windows)

Wireless Networks | WLAN

WLAN connectivity standards

- 801.11 specs define the operating mode for the MAC sublevel and protocols for the physical layer (e.g., IR – InfraRed , DSSS - Direct Sequence Spread Spectrum, FHSS - Frequency Hopping Spread Spectrum)



Wireless Networks | WLAN

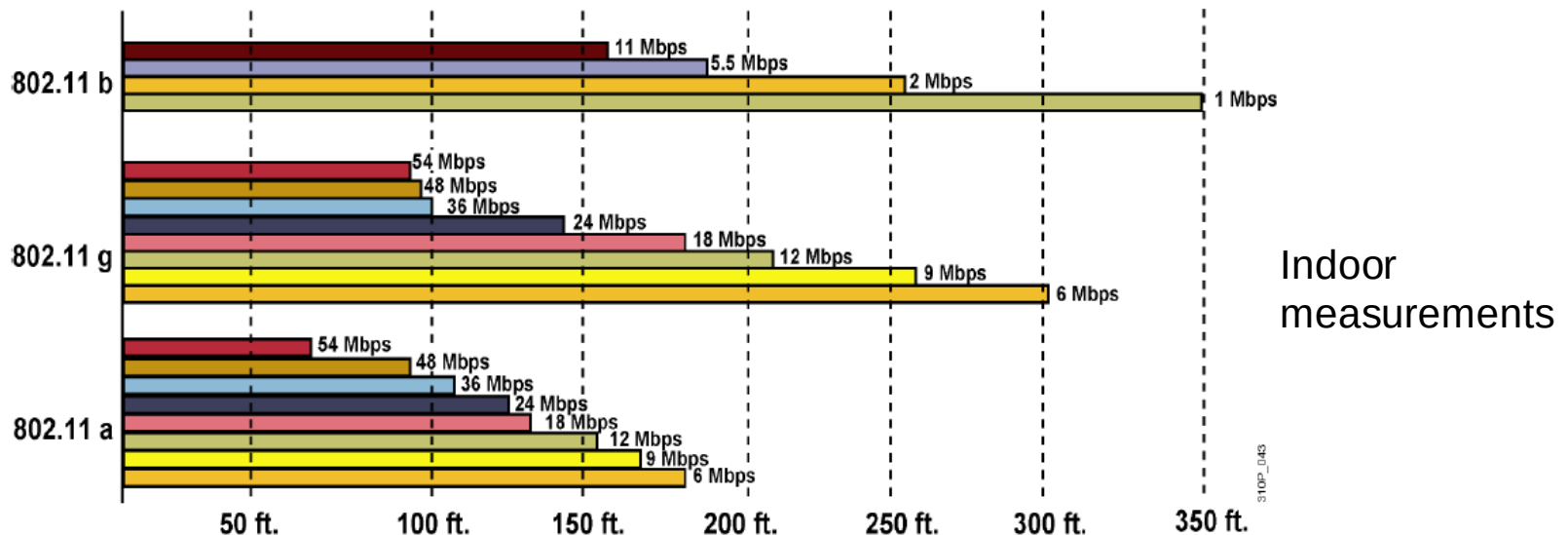
WLAN connectivity standards

- **IEEE 802.11b** (since 1999)
 - Uses 2.4 Ghz frequency
 - The *throughput* can be 11 Mbps maximum (effective: 5Mbps; DSSS and FHSS)
 - Covered area ~100m (transfer speed depends on the distance to the device; e.g., for a distance of 65m between two devices, the transfer is done at a maximum of 2 Mbps)
- **IEEE 802.11a**
 - Uses 5 Ghz frequency (less collisions)
 - Offers up to 54 Mbps (effective: 11-18 Mbps)
 - Requires higher transmission power and a smaller area than 802.11b (more APs may be needed)
 - It is not compatible with 802.11b/g

Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11g**
 - Standard since 2003, combines various features from the previous
 - Compatible with 802.11b
 - Uses 2.4 Ghz frequency
 - Increases throughput to 54Mbps (effective: 20 - 25Mbps)



Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11n-2009**

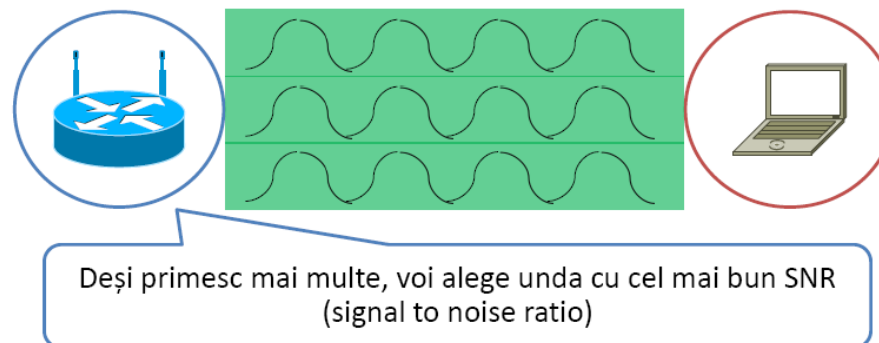
- Uses 2.4 Ghz or 5Ghz
- Significantly improves data transfer from 802.11a, 802.11g - 54 Mbit/s to 600 Mbit/s => a threat to FastEthernet and a possible platform for telephony and video signals
- Compatible with previous standards
- Uses multiple antennas and MIMO technology
- Increased immunity to noise using advanced carrier wave modulation
- Supports packet aggregation (a single header for multiple data packets)

Wireless Networks | WLAN

WLAN connectivity standards

MIMO (SU-MIMO or Single User MIMO)

- MIMO uses advanced processors for spatial multiplexing of the signal
- The processor can multiplex and demultiplex data in order to obtain a higher throughput
- multipath effect = the process in which multiple carrying waves of the same information are received, but who reflected differently and with a different clarity
- In 802.11g, the processor chooses the clearest wave

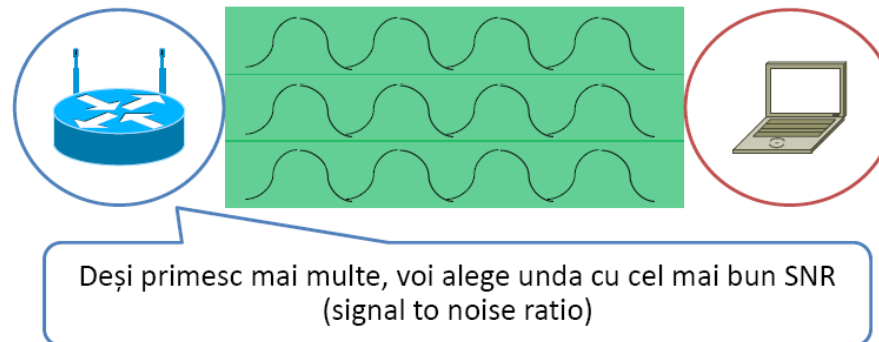


Wireless Networks | WLAN

WLAN connectivity standards

MIMO (SU-MIMO or Single User MIMO)

- Problem: waves with weaker SNR are ignored, even if they may contain relevant information
- In 802.11n, the MRC (Maxim Ratio Combining) technology implemented in the network card's processor composes all waves in order to obtain the maximum clarity of the information => better throughput
- Example: If you have an 802.11n card in an 802.11g network, you will have a better throughput than 802.11g clients



Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11ad - WiGig**
 - Standard announced in 2009
 - 2011 – version 1.1
 - Uses 60 GHz bands
 - Transfer rate of aprox. 7 Gbit/s (10 times faster than the maximum rate of 802.11n)
 - Transmission distance up to 10m
 - Maintains compatibility with the existing standards
- **IEEE 802.11ay – WiGig**
 - Improvement of 802.11ad
 - The development started in 2015
 - The standard was approved in 2021
 - Transfer rate between 20-40Gbit/s (without channel bonding)
 - Transmission distance up to 300-500m



Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11ac**
 - Developed since 2008; December 2013 – standard
 - Uses 5 GHz band
 - **MIMO** (multiple-input and multiple-output) → **MU-MIMO** (Multi-user MIMO) algorithms are developed to enhance MIMO systems when the number of users or connections is greater than one
 - Theoretical maximum speed of 1.3–2.3 Gbps (can be more, depending on the number of supported spatial streams)



Standard Wi-Fi

Tehnologie

802.11 b

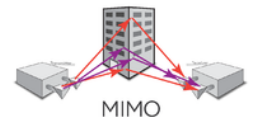
802.11 a

802.11 g

802.11 n

802.11 ac

MU-MIMO



Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11ax**
 - Successor of 802.11ac
 - Officially standardized in 2021
 - Known as Wi-Fi 6(E)
 - Uses 2.4 GHz, 5 GHz and 6 GHz frequency bands
 - Theoretical max. speed of approx. 14 Gbps (depends on the number of supported spatial streams)
 - Support for MU-MIMO
 - Introduces support for OFDMA (Orthogonal Frequency Division Multiple Access), which complements MU-MIMO
 - Compatibility with previous standards

Wireless Networks | WLAN

WLAN connectivity standards

- **IEEE 802.11be**
 - Successor of 802.11ax
 - Released in 2024
 - Known as Wi-Fi 7
 - Uses 2.4 GHz, 5 GHz and 6 GHz frequency bands
 - Theoretical max. speed of approx. 46 Gbps
 - Up to 4x lower latency than Wi-Fi 6(E)
 - Support for MU-MIMO
 - Enhanced resource allocation in OFDMA
 - Compatibility with previous standards

Wireless Networks | WLAN

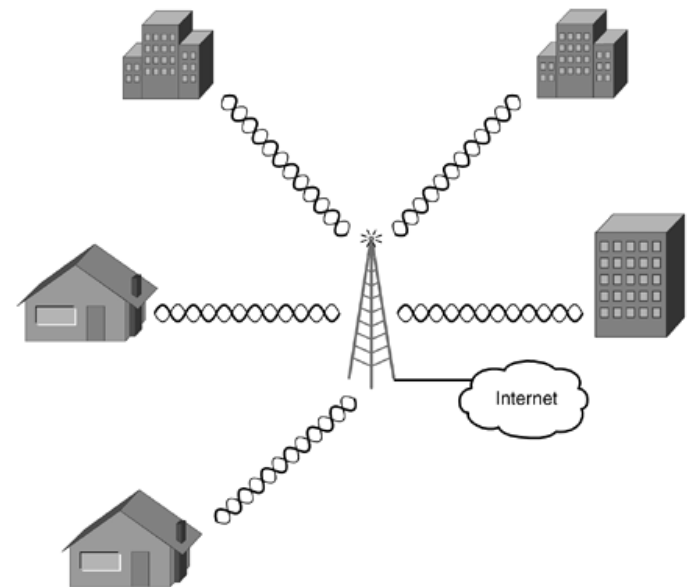
WLAN connectivity standards

- **HomeRF (Radio Frequency) (discontinued)**
 - Since: 1998 -> 2003, initiated by Siemens, Motorola, Philips
 - For home communications
 - Supports quality voice communications
- **HiperLAN/2 (High Performance Radio LAN) (discontinued)**
 - European alternative to IEEE 802.11 standards - European Telecommunications Standards Institute (ETSI)
 - Uses 5Ghz band
 - Maximum transfer of 54 Mbps, on a radius of ~150m
 - Incorporates features for QoS provisioning (for real-time multimedia transmissions)

Wireless Networks | WMAN

WMAN (*Wireless Metropolitan-Area Network*)

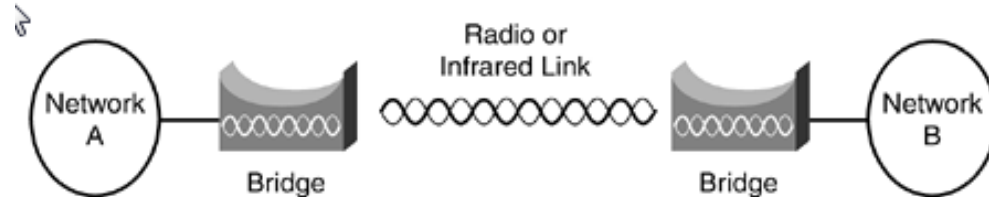
- Operating space: a city
- In general, there is a fixed layout of the network
- Useful when the traditional wired connections are not feasible
- Standards:
 - 802.16
 - 802.11 is also used



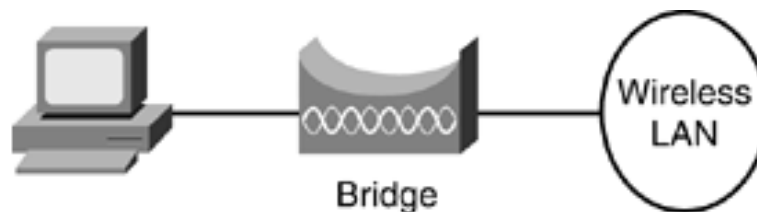
Wireless Networks | WMAN

WMAN - components

- **Bridges:** ensure connectivity between two networks which use similar or different protocols in data link layer



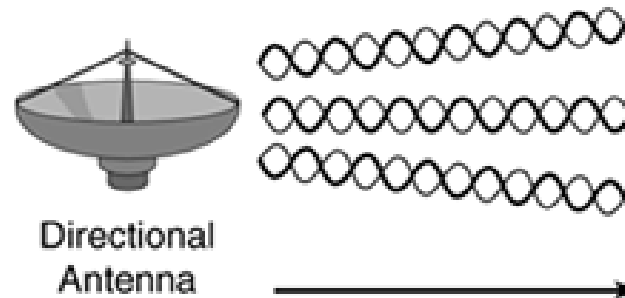
- Are different from an access point
- An *Ethernet-Wireless bridge* can be used for connecting through an Ethernet port a device to an access point



Wireless Networks | WMAN

WMAN - components

- **Antennas:** usually, directional antennas are used for WMAN networks, to maximize the intensity of radio waves in a single direction



- Other types of antennas can be used (e.g., semi-directional antennas cover a double surface than omnidirectional ones)

Wireless Networks | WMAN

WMAN systems – provide connectivity between buildings and users within a city using a few configurations

- *point-to-point*: uses RF or infrared transmission, using (semi)directional antennas that can reach 30 miles for RF systems



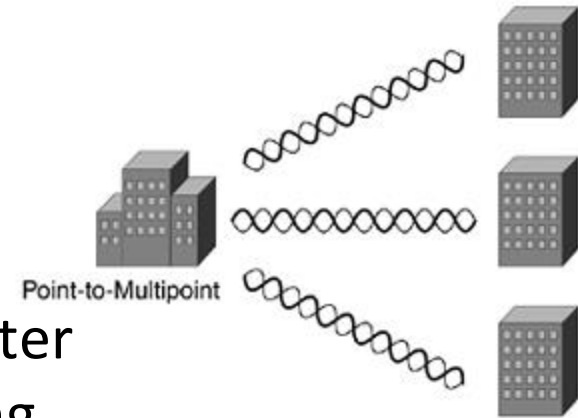
Example: a medical center can use a point-to-point communication between the main hospital and a clinic in the same city

Wireless Networks | WMAN

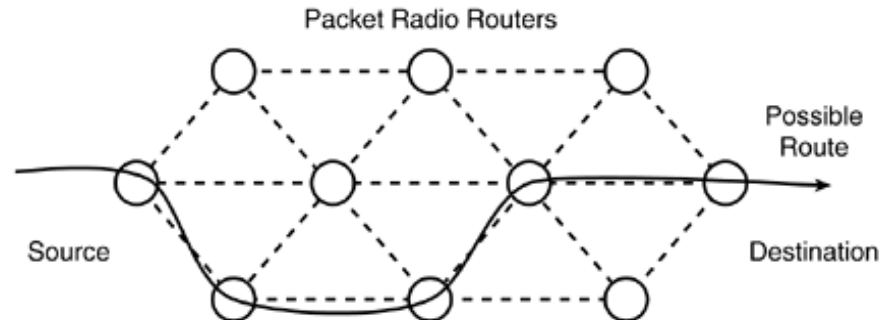
WMAN systems

- *point-to-multipoint*: use an omnidirectional antenna

Example: a company with a headquarter and warehouses and/or manufacturing facilities in the same city or rural area



- “*packet radio*” systems: use special wireless routers that transmit packets between them



Wireless Networks | WMAN

WMAN connectivity standards

- **WI-FI/802.11**
 - Directional antennas and wireless bridges are used
 - Low costs, but the performance is limited in case of a large number of users
 - RF interferences appear if there are many 802.11 networks
- **802.16** or **WiMAX** (Worldwide Interoperability for Microwave Access)
 - Example:
 - 802.16-2017 IEEE Standard for Air Interface for Broadband Wireless Access Systems

Wireless Networks | WMAN

WMAN connectivity standards

- **802.16** sau **WiMAX** (Worldwide Interoperability for Microwave Access)

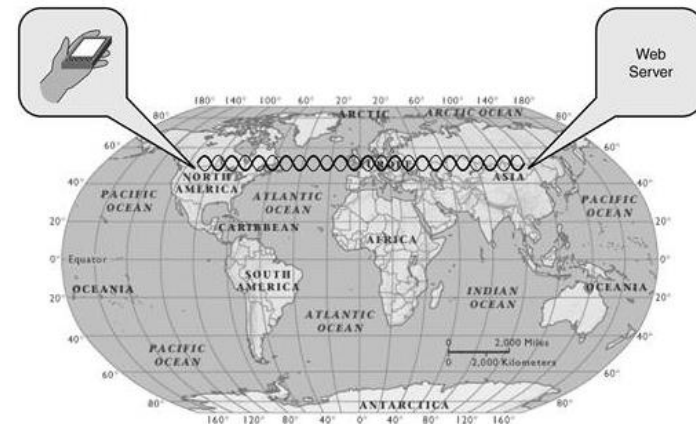
802.16.2	IEEE Recommended Practice for Local and metropolitan area networks Coexistence of Fixed Broadband Wireless Access Systems (Maintenance and rollup of 802.16.2–2001 and P802.16.2a) Released on 2004-March-17.	Current
802.16k	IEEE Standard for Local and Metropolitan Area Networks: Media Access Control (MAC) Bridges Amendment 2: Bridging of IEEE 802.16 (An amendment to IEEE 802.1D) Released on 2007-August-14.	Current
802.16.1	IEEE Standard for WirelessMAN-Advanced Air Interface for Broadband Wireless Access Systems Released on 2012-September-07.	Current
802.16p	IEEE Standard for Air Interface for Broadband Wireless Access Systems Amendment 1: Enhancements to Support Machine-to-Machine Applications Released on 2012-October-08.	Current
802.16.1b	IEEE Standard for WirelessMAN-Advanced Air Interface for Broadband Wireless Access Systems Amendment 1: Enhancements to Support Machine-to-Machine Applications Released on 2012-October-10.	Current
802.16n	IEEE Standard for Air Interface for Broadband Wireless Access Systems Amendment 2: Higher Reliability Networks Approved on 2013-March-06.	Current
802.16.1a	IEEE Standard for WirelessMAN-Advanced Air Interface for Broadband Wireless Access Systems Amendment 2: Higher Reliability Networks Approved on 2013-March-06.	Current
802.16-2017	IEEE Standard for Air Interface for Broadband Wireless Access Systems It is a rollup of 802.16p, 802.16n, 802.16q and Std 802.16s Released on 2017-September.	Current

Wireless Networks | WWAN

WWAN (*Wireless Wide-Area Network*)

- Operating space: globally, as a result of the cooperation of several telecommunications companies (AT&T, Verizon, Orange, etc.)
- Data transfer is done by packet switching
 - In contrast with virtual circuit switching
 - It does not require dedicated connections
 - Allow multiple users to use a single connection

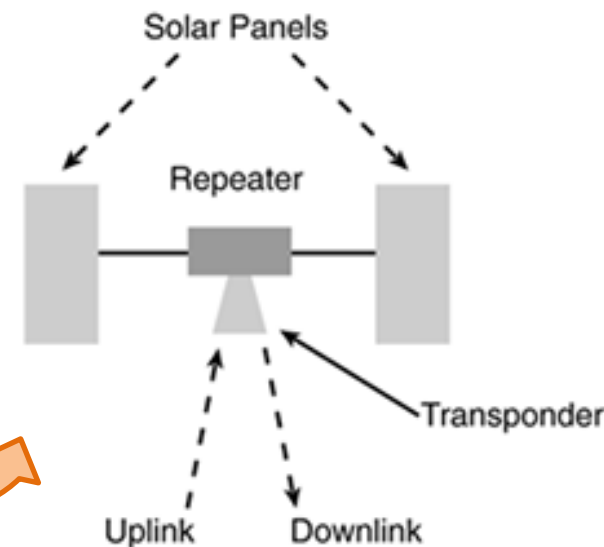
[Wireless Networks first-step,
Jim Geier]



Wireless Networks | WWAN

WWAN – components

- **Radio NIC** – useful for integrating a laptop or other devices in a WWAN; some mobile phones contain such cards, but integration issues exist because there are different types of WWANs (e.g., depending on the satellite hardware)
- **Base Stations**
 - *Cell towers*
 - *Sattellites*
- **Antennas**

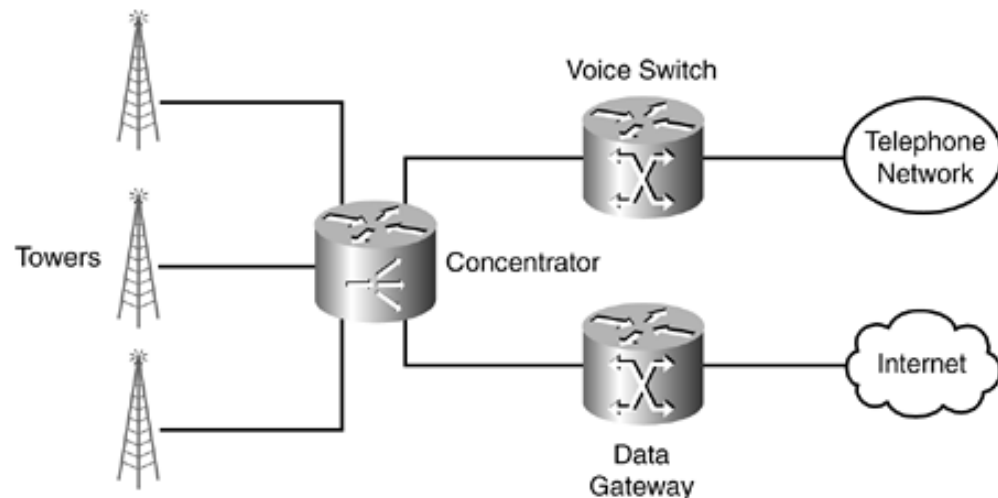


Wireless Networks | WWAN

WWAN systems

WWAN based on cells

- Consisting of:
 - *Cell towers*: receive signals from a user and transmit information back to him
 - *Voice switches* – connects the device of the user to another user using the telephone system
 - *Gateways* – turn the system into a WWAN; makes it possible for users to access the Internet



[Wireless Networks first-step,
Jim Geier]

Wireless Networks | WWAN

WWAN systems

WWAN based on cells

- The cell: the geographical area which is covered by the signal
 - Depends on the protocol, signal strength, obstacles
 - Range of action 1-40km
 - The cells are coordinated by a cell system
 - Very populated areas contain micro-cells (diam. ~100m)
 - The connection between users in different cells is passed through a process called *handoff* or *handover*
 - The placement of the network equipment is done according to optimization strategies of the coverage and maximization strategies of the signal

Wireless Networks | WWAN

WWAN

- **Multiplexing:** the signal is used (shared) by multiple users
 - **Frequency-division multiple access (FDMA):** each signal within the communication channel has a unique frequency (the radio station model)
 - **Time-division multiple access (TDMA):** assigns to each user segments of time when he can communicate
 - **Code-division multiple access (CDMA):** each signal has a code attached, all signals being transmitted in order to “fill” the entire bandwidth; the receiver will only process signals having the correct code
 - **Orthogonal frequency-division multiple access (OFDMA):** divides a channel into smaller frequency allocations, named Resource Units (RUs); the communication with multiple clients is done by assigning them to specific RUs

Wireless Networks | WWAN

WWAN systems

WWAN 2nd generation (2G)

- Improvement over first generation ('70-'80)
- Besides digital voice transmission, it supports data transmissions (9.6 – 19.2 Kbps)
- Code Division Multiple Access (CDMA IS-95)
- **Global System for Mobile Communication (GSM)**
- Features: SMS, calendar applications, personal information management (PIM), tones, games, WAP access...

WAP

WAP (Wireless Application Protocol)

- Standard allowing access to information and services provided through the Internet via a mobile device
 - under the coordination of Open Mobile Alliance (former WAP Forum)

www.wapforum.org

- Versions: WAP1.0, WAP 1.1, WAP 1.2, WAP 2.0

WAP

WAP (Wireless Application Protocol)



- **WAP Gateway:**
mediator between
Internet and a
device using WAP

For WAP 2, it is not
necessary to use,
HTTP/1.1 being
used directly

[International Engineering Consortium,
<http://www.iec.org>]

Wireless Networks | WWAN

WWAN systems

WWAN generation 2.5 (2.5G):

- Generally, the transfer rate is ~115kbps
- **General Packet Radio Services (GPRS)** – max. 171.2 kbps ->
Enhanced Data for Global Evolution (EDGE) – max. 474 kbps
- Code Division Multiple Access CDMA 2000 1x
- GPRS networks behave similarly to classic LANs, making applications easier to implement
- A GPRS device <-> terminal

Wireless Networks | WWAN

WWAN systems

WWAN generation 3 (3G):

- The 3GPP (Third Generation Partnership Project) family of standards defined by the International Telecommunication Union (www.3gpp.org, www.3gpp2.org)
- It is based on Universal Mobile Telecommunications System (UMTS) (Wideband CDMA (W-CDMA), TD-CDMA (Time Division Code Division Multiple Access), ...)
- Transfer rate
 - 128 Kbits/s for fast-moving devices
 - 384 Kbits/s for slow-moving devices
 - 2.05 Mbits/s for stationary devices

Wireless Networks | WWAN

WWAN systems

WWAN generation 3 (3G):

- Features of 3G services:
 - Increased mobility and connectivity (3G networks use IP)
 - Provide multimedia services (audio and video streaming)
 - Fast download of large files
 - Access to corporate applications
 - Provide QoS
- Discussions: 3G alternative to WLAN
 - WLAN – offers a higher transfer rate
 - Developing a WLAN is less costly

Wireless Networks | WWAN

WWAN systems

WWAN generation 4 (4G):

- Transfer rates up to 1 Gbit/s (Gigabit LTE)
- A major difference from 3G, which is based on two parallel infrastructures (packet switched and circuit switched), is that it is based only on a packet switched infrastructure
- Uses Voice over LTE (VoLTE) standard
- Uses OFDMA
- Smaller latency than 3G
- **Pre-4G (3G Extensions): 3GPP**
 - 3GPP LTE (Long Term Evolution) Standard
 - Launched in december 2009, Telia Sonera – first “4G” services in Sweden and Norway (user terminals – Samsung)
 - IEEE 802.16m or WirelessMAN-Advanced or WiMAX
 - UMB (Ultra Mobile Broadband) or 3GPP2
- 6 December 2010 ITU (International Communication Union) -> current versions (LTE, UMB, ...) do not comply with the 4G standard

Wireless Networks | WWAN

WWAN systems

WWAN generation 5 (5G):

- Transfer rates up to 20 Gbit/s
- Uses a packet switched infrastructure
- Uses Voice over LTE (VoLTE) standard
- Uses OFDMA
- Even lower latency than 4G
- Less interferences, which leads to less congestion
- Less used energy for mobile devices
- The range is influenced by many factors, the used frequency being the most important

<https://www.qualcomm.com/5g/what-is-5g>

Wireless Internet

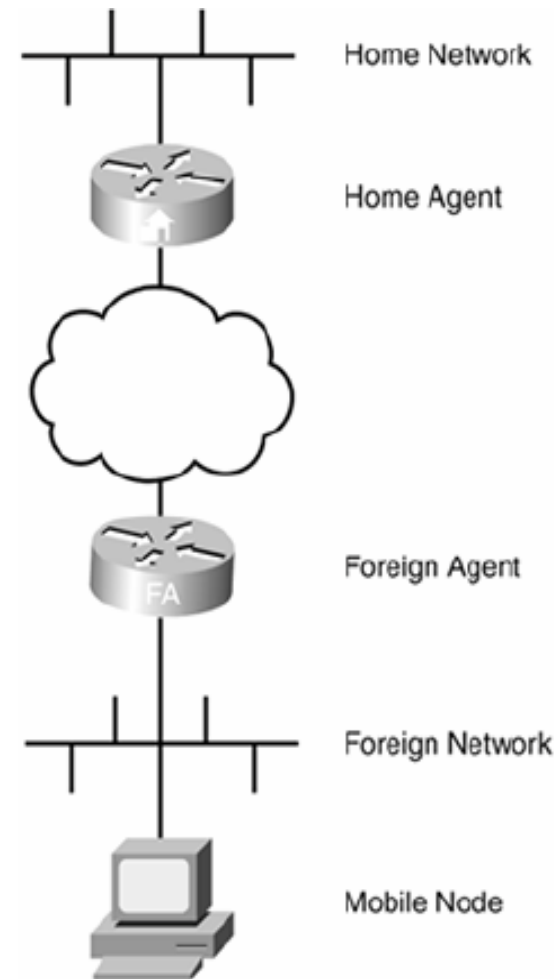
- **Mobile IP**

- Protocol that allows a mobile device to move from one network to another and maintain its IP address
 - Mobile IPv4 – RFC 5944, RFC 4721
 - Mobile IPv6 – RFC 6275, RFC 6543 & 7864 (Proxy Mobile IPv6 – PMIPv6)
 - Routing protocol in which end devices signal their own route updates and dynamic data tunnels eliminate the need to propagate route information
 - An user can use (roam) various IP subnetworks and access links, maintaining a continuous communication
- Extension:
 - Network Mobility (NEMO) - RFC 3963, RFC 6626

Wireless Internet: Mobile IP

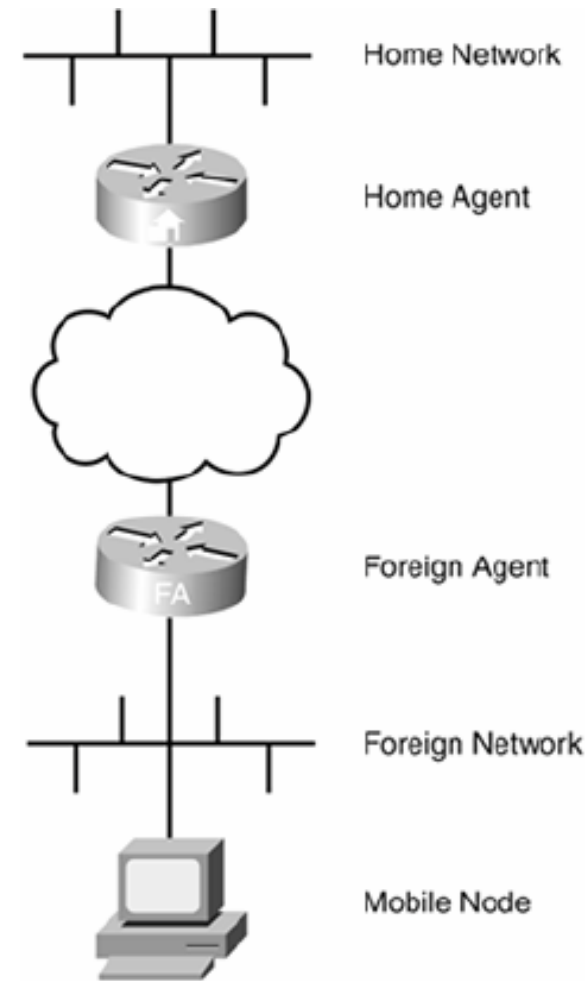
- Components:

- **Mobile Node**: a device using IP
- **Home Agent (HA)**: node responsible for redirecting data to the current location of the mobile node
 - Processes IP routing updates (registrations)
 - Send data via dynamic tunnels
- **Foreign Agent (FA)**: a router attached to an access link, located at the other end of the tunnel established with a mobile node
 - Offers (advertises) one or more IP addresses referred to as **CoA** (Care-of Address)
 - When a node registers to a Home Agent, it does it via an FA
 - An FA must be directly connected to the mobile node



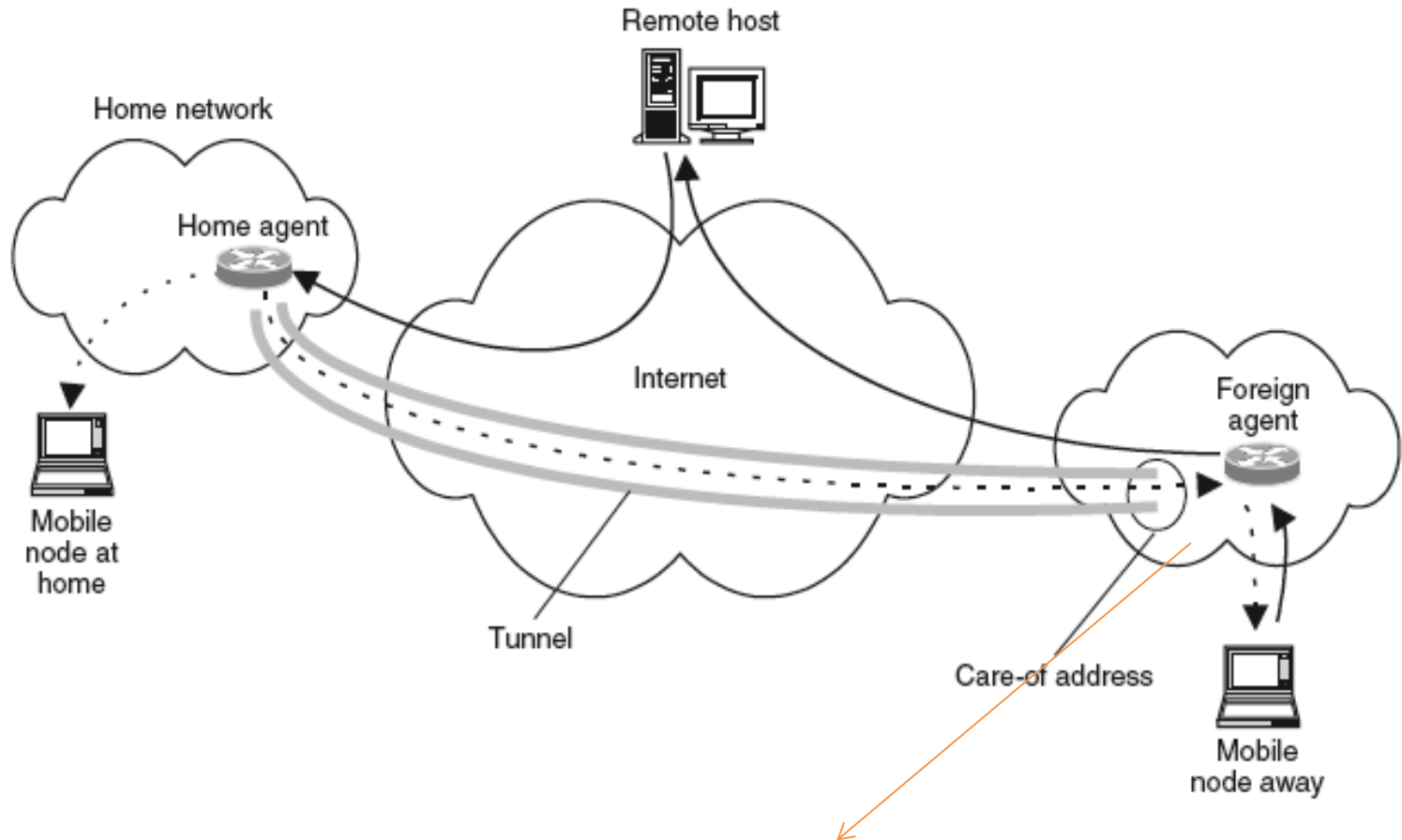
Wireless Internet: Mobile IP

- The mobile node has two addresses:
 - **Home Address**
 - The IP address of the mobile node;
 - Can be statically or dynamically assigned, during registration
 - **CoA (*Care-of Address*)**
 - Valid and routable IP address
 - Defines the edge of the network that can be accessed through standard routing
 - Represents a tunnel's terminal point



[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Wireless Internet: Mobile IP



If the **mobile node** is in the *Foreign Network*, the messages are sent to the *Home agent*, then they are sent through the tunnel to the *Foreign agent*, which will send them to the **mobile node**

[Wireless Networking Complete,
Morgan Kaufmann]

Wireless Internet: Mobile IP

General mechanism:

A node wishing to communicate with the mobile node will use its permanent home address

- If the mobile node is found in *HomeNetwork*, then if someone sends a packet, *IP forwarding* is used
- If the mobile node is found in a foreign network, CoA is used
 - The mobile node registers its real location to the *home agent*
 - The packets are sent through a tunnel from the *home agent* to CoA (end of the tunnel)

Obs.: If the node does not receive messages of type *agent advertisement*, then it tries to obtain an address through techniques like DHCP to know its current location

Wireless Internet: Mobile IP

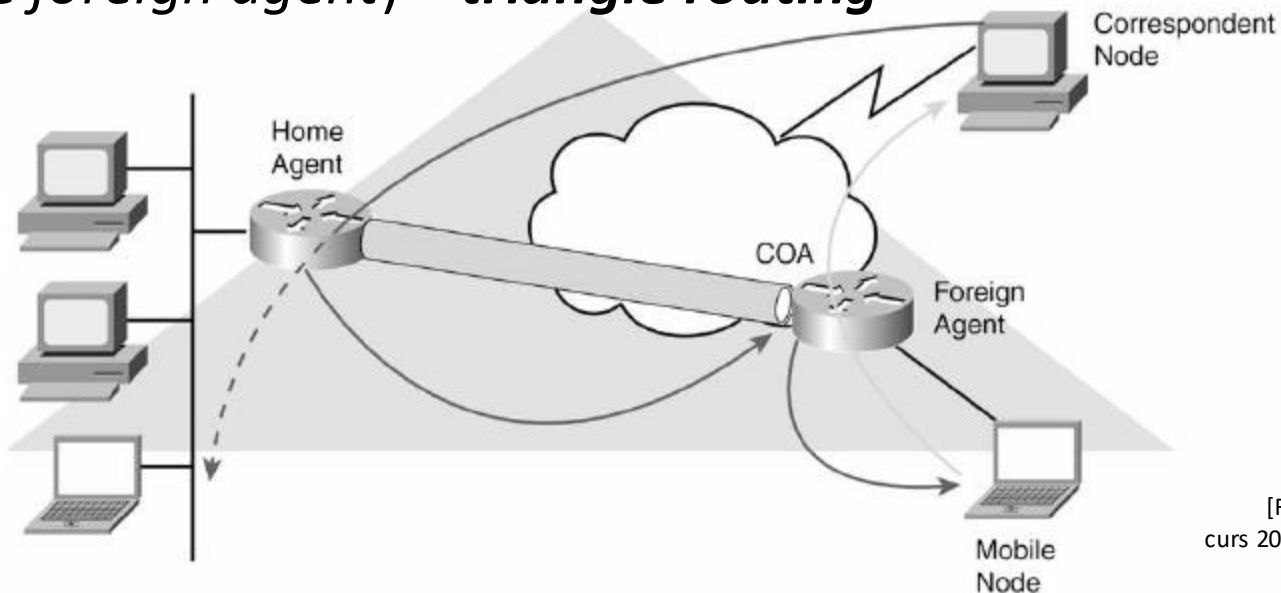
Tunnels: logical links at a distance of 1 hop, located at the edges of Foreign Network to which the mobile nodes are attached

- Can transport any IP packets between the endpoints of the communication
- Data encapsulation is done via IP-in-IP – RFC 2003 (20 additional bytes)

Wireless Internet: Mobile IP

Routing:

- When the **mobile node** wants to send packets, it will not send them to the *home agent*, it will send them directly to the node with whom it wants to communicate, using *home address* as value for the *source address* field of the IP packet (using of course *foreign agent*) – **triangle routing**



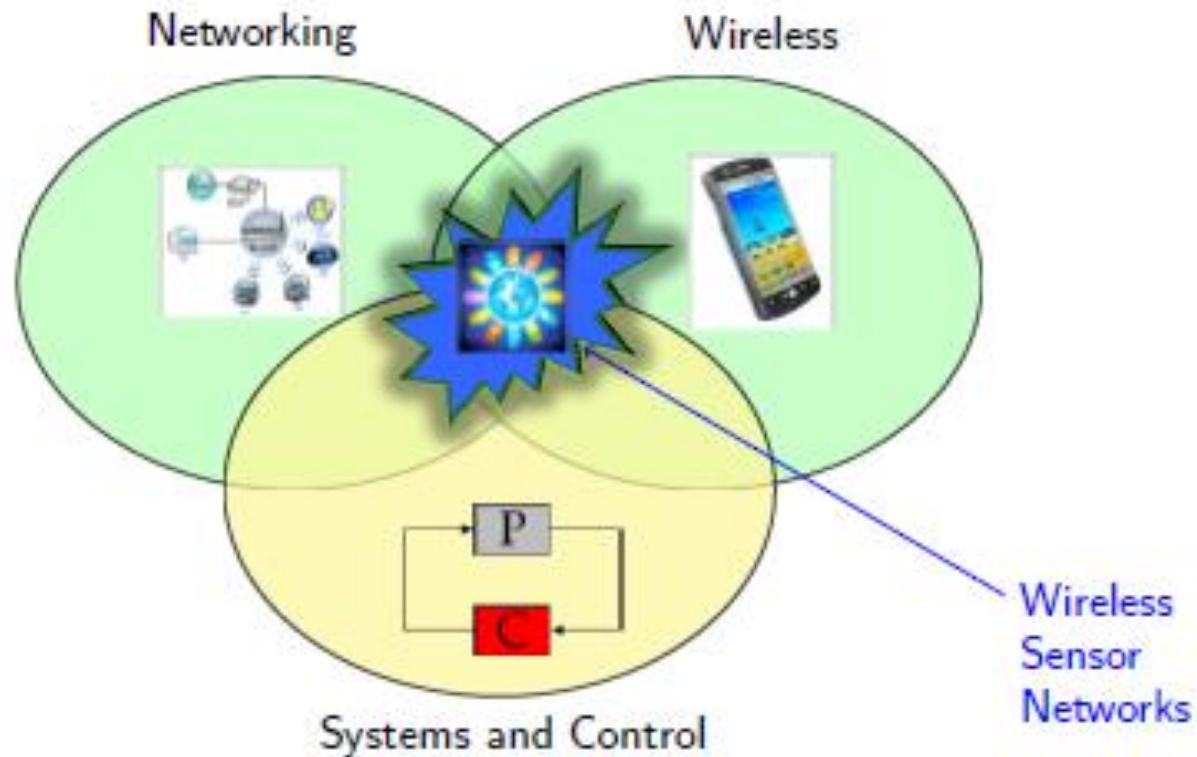
[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Wireless Applications

- Wireless networks provide support in many fields
 - Internet access
 - *Voice over Wireless*
 - Healthcare
 - Education
 - Services
 - Sales
 - ...

Wireless Applications

Wireless Sensor Network



Wireless Applications

- “*Wearable sensors*”



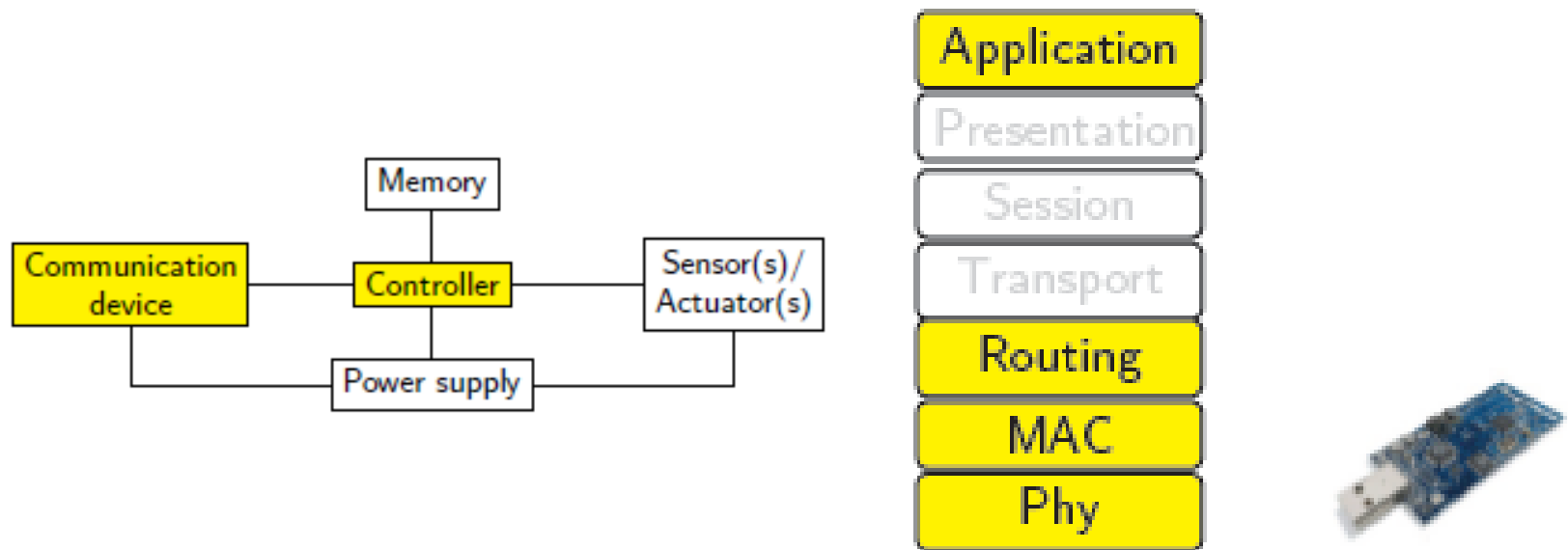
- “*Smart buildings*”



- “*Environmental monitoring*”: the degree of water pollution

WSN | Components of a wireless node

The behavior of a node is specified by a set of protocols



[Cosmin Varlan ...Curs...Arduino...]

Wireless Applications

Challenges:

- Provide continuous connectivity
 - Quality of content
 - Concurrency (sharing resources/resource management)
 - Agility (speed/flexibility in adapting to changes)
- Reducing the consumed power
- Ensure device independency of the wireless applications

Bibliography

- Network+ Guide to Networks, Fifth Edition, Tamara Dean, Network +, ISBN-13: 978-1-423-90245-4, 2009
- Wireless Networks first-step, Jim Geier, Publisher: Cisco Press, 2004
- <https://www.amazon.com/Building-Wireless-Sensor-Networks-Processing/dp/0596807732>
- <https://smallbusiness.chron.com/bluetooth-mouse-vs-wireless-mouse-55689.html>
- <https://www.bluetooth.com/>
- <https://www.makeuseof.com/tag/top-uses-for-bluetooth-on-your-android-phone/>
- Is Bluetooth Radiation Really Dangerous? URL: <https://www.rfsafe.com/bluetooth-radiation-dangerous-cell-phone-radiation/>
- <https://www.cnet.com/tech/mobile/att-jumps-into-fast-lane-with-ultraspeedy-5g-field-trial/>
- <https://books.google.ro/books?id=UcL1CwAAQBAJ&pg=PA25&lpg=PA25&dq=wwan+5G&source=bl&ots=9ENJYCAu-R&sig=uyXVSg9rDy63clQBSeVETGSEdSk&hl=en&sa=X&ved=0ahUKEwi575up-bXRAhXCJSwKHSYsDcwQ6AEINjAE#v=onepage&q&f=false>
- <https://www.revsys.com/writings/quicktips/ssh-tunnel.html>

Summary

- Preliminaries
- Components
- Types of wireless networks. Characteristics.
- Mobile IP
- Applications
- Wireless Sensor Networks (WSN)



Questions?

Questions?